Case Brief: Akiba SACCO

Feature Phone Web Emulator

Background

Akiba SACCO is upgrading its USSD self-service portal for members using legacy feature phones. Before rolling out changes to the live telco gateway, the customer care and operations teams require a browser-based emulator to test menu depth, train support agents, and preview text constraints.

You are tasked with designing the interface in Figma and implementing a functional vanilla web prototype.

Product Specifications

1. Hardware & Display Constraints

- **Target Handset:** Basic feature phone with physical keypad navigation and a fixed monochrome screen.
- **Display Area:** The screen display is strictly **160px wide by 128px tall**. Text must fit within this boundary without scrolling or overflowing.
- **Input Device:** All user actions must originate from on-screen keypad buttons (`0–9`, `*`, `#`, and optional Call/End keys). No direct clicks or touches are allowed inside the display screen area.

2. User Journeys & Menu Architecture

The emulator must support two core transaction paths starting from the dial code `*144#`:

```
[Dial *144#]
     |
[Main Menu] ── 1. Balance Enquiry ──> [Balance Display]
     |
     └── 2. Withdraw Money ──> [Withdraw Menu]
                                   ├── 1. My Number    ──> [Enter Amount] ──> [Confirmation]
                                   └── 2. Other Number ──> [Enter Phone]  ──> [Enter Amount] ──> [Confirmation]
```

3. Localization

Akiba SACCO operates in bilingual (English, Swahili) markets. Every screen state in your Figma file must include both language variants side-by-side

Deliverables

Phase 1: Figma Prototype

- Set all screen frames to exactly **160×128 px**.
- Design the handset casing, physical buttons, and monochrome display.
- Map out the complete interaction tree (Main Menu, Sub-menus, Input Prompts, Confirmations) for both English and Swahili.
- Connect interactive nodes using Figma Prototype mode to demonstrate navigation and back-button routing.

Phase 2: Web Implementation

Deliver a zero-dependency web build

- **Fidelity:** Replicate your Figma handset casing, keypad layout, typography, and color palette.
- **Structure & Layout:** Use HTML and CSS for the structure and layout. Screen elements must strictly respect the 160×128px constraint.
- **Language:** Use TypeScript for all application logic.